

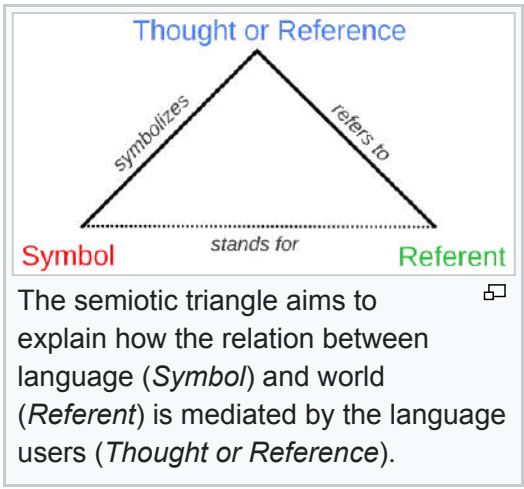
Truth is a property of statements that accurately present the world and true statements are in accord with **reality**. Whether a statement is true usually depends on the relation between the statement and the rest of the world. The **truth conditions** of a statement are the way the world needs to be for the statement to be true. For example, it belongs to the truth conditions of the sentence "it is raining outside" that raindrops are falling from the sky. The sentence is true if it is used in a situation in which the truth conditions are fulfilled, i.e., if there is actually rain outside.^[39]

Truth conditions play a central role in semantics and some theories rely exclusively on truth conditions to analyze meaning. To understand a statement usually implies that one has an idea about the conditions under which it would be true. This can happen even if one does not know whether the conditions are fulfilled.^[39]

Semiotic triangle [[edit](#)]

The **semiotic triangle**, also called the triangle of meaning, is a model used to explain the relation between language, language users, and the world, represented in the model as *Symbol*, *Thought or Reference*, and *Referent*. The symbol is a linguistic **signifier**, either in its spoken or written form. The central idea of the model is that there is no direct relation between a linguistic expression and what it refers to, as was assumed by earlier dyadic models. This is expressed in the diagram by the dotted line between symbol and referent.^[40]

The model holds instead that the relation between the two is mediated through a third component. For example, the term *apple* stands for a type of fruit but there is no direct connection between this string of letters and the corresponding physical object. The relation is only established indirectly through the mind of the language user. When they see the symbol, it evokes a mental image or a concept, which establishes the connection to the physical object. This process is only possible if the language user learned the meaning of the symbol before. The meaning of a specific symbol is governed by the conventions of a particular language. The same symbol may refer to one object in one language, to another object in a different language, and to no object in another language. [[40](#)]



Others [[edit](#)]

Many other concepts are used to describe semantic phenomena. The **semantic role** of an expression is the function it fulfills in a sentence. In the sentence "the boy kicked the ball", the boy has the role of the agent who performs an action. The ball is the theme or patient of this action as something that does not act itself but is involved in or affected by the action. The same entity can be both agent and patient, like when someone cuts themselves. An entity has the semantic role of an instrument if it is used to perform the action, for instance, when cutting something with a knife then the knife is the instrument. For some sentences, no action is described but an experience takes place, like when a girl sees a bird. In this case, the girl has the role of the experiencer. Other common semantic roles are location, source, goal, beneficiary, and stimulus.^[41]

Lexical relations describe how words stand to one another. Two words are **synonyms** if they share the same or a very similar meaning, like *car* and *automobile* or *buy* and *purchase*. **Antonyms** have opposite meanings, such as the contrast between *alive* and *dead* or *fast* and *slow*.^[c] One term is a **hyponym** of another term if the meaning of the first term is included in the meaning of the second term. For example, *ant* is a hyponym of *insect*. A **prototype** is a hyponym that has characteristic features of the type it belongs to. A *robin* is a prototype of a *bird* but a *penguin* is not. Two words with the same pronunciation are **homophones** like *flour* and *flower*, while two words with the same spelling are **homonyms**, like a bank of a river in contrast to a bank as a financial institution.^[d] Hyponymy is closely related to **meronymy**, which describes the relation between part and whole. For instance, *wheel* is a meronym of *car*.^[44] An expression is **ambiguous** if it has more than one possible meaning. In some cases, it is possible to **disambiguate** them to discern the intended meaning.^[45] The term ***polysemy*** is used if the different meanings are closely related to one another, like the meanings of the word *head*, which can refer to the topmost part of the human body or the top-ranking person in an organization.^[44]

The meaning of words can often be subdivided into meaning components called **semantic features**. The word *horse* has the semantic feature *animate* but lacks the semantic feature *human*. It may not always be possible to fully reconstruct the meaning of a word by identifying all its semantic features.^[46]

A **semantic** or lexical field is a group of words that are all related to the same activity or subject. For instance, the semantic field of **cooking** includes words like *bake*, *boil*, *spice*, and *pan*.^[47]

The **context** of an expression refers to the situation or circumstances in which it is used and includes time, location, speaker, and audience. It also encompasses other passages in a text that come before and after it.^[48] Context affects the meaning of various expressions, like the **deictic** expression *here* and the **anaphoric** expression *she*.^[49]

A syntactic environment is **extensional or transparent** if it is always possible to exchange expressions with the same reference without affecting the truth value of the sentence. For example, the environment of the sentence "the number 8 is even" is extensional because replacing the expression "the number 8" with "the number of planets in the **Solar System**" does not change its truth value. For **intensional or opaque contexts**, this type of substitution is not always possible. For instance, the **embedded clause** in "Paco believes that the number 8 is even" is intensional since Paco may not know that the number of planets in the solar system is 8.^[50]

Semanticists commonly distinguish the language they study, called object language, from the language they use to express their findings, called **metalanguage**. When a professor uses Japanese to teach their student how to interpret the language of **first-order logic** then the language of first-order logic is the object language and Japanese is the metalanguage. The same language may occupy the role of object language and metalanguage at the same time. This is the case in **monolingual English dictionaries**, in which both the entry term belonging to the object language and the definition text belonging to the metalanguage are taken from the English language.^[51]

Branches [[edit](#)]

Lexical semantics [[edit](#)]

*Main article: **Lexical semantics***

Lexical semantics is the sub-field of semantics that studies word meaning.^[52] It examines semantic aspects of individual words and the **vocabulary** as a whole. This includes the study of lexical relations between words, such as whether two terms are synonyms or antonyms.^[53] Lexical semantics categorizes words based on semantic features they share and groups them into semantic fields unified by a common subject.^[54] This information is used to create taxonomies to organize lexical knowledge, for example, by distinguishing between physical and **abstract entities** and subdividing physical entities into **stuff** and **individuated entities**.^[55] Further topics of interest are polysemy, ambiguity, and **vagueness**.^[56]

Lexical semantics is sometimes divided into two complementary approaches: **semasiology** and **onomasiology**. Semasiology starts from words and examines what their meaning is. It is interested in whether words have one or several meanings and how those meanings are related to one another. Instead of going from word to meaning, onomasiology goes from meaning to word. It starts with a concept and examines what names this concept has or how it can be expressed in a particular language.^[57]

Some semanticists also include the study of lexical units other than words in the field of lexical semantics. **Compound expressions** like *being under the weather* have a non-literal meaning that acts as a unit and is not a direct function of its parts. Another topic concerns the meaning of morphemes that make up words, for instance, how negative **prefixes** like *in-* and *dis-* affect the meaning of the words they are part of, as in *inanimate* and *dishonest*.^[58]

Phrasal semantics [[edit](#)]

Phrasal semantics studies the meaning of sentences. It relies on the principle of compositionality to explore how the meaning of complex expressions arises from the combination of their parts.^{[59][e]} The different parts can be analyzed as **subject**, **predicate**, or **argument**. The subject of a sentence usually refers to a specific entity while the predicate describes a feature of the subject or an event in which the subject participates. Arguments provide additional information to complete the predicate.^[61] For example, in the sentence "Mary hit the ball", *Mary* is the subject, *hit* is the predicate, and *the ball* is an argument.^[61] A more fine-grained categorization distinguishes between different semantic roles of words, such as agent, patient, theme, location, source, and goal.^[62]

Verbs usually function as predicates and often help to establish connections between different expressions to form a more complex meaning structure. In the expression "Beethoven likes Schubert", the verb *like* connects a liker to the object of their liking.^[63] Other sentence parts modify meaning rather than form new connections. For instance, the **adjective** *red* modifies the color of another entity in the expression *red car*.^[64] A further compositional device is variable binding, which is used to determine the reference of a term. For example, the last part of the expression "the woman who likes Beethoven" specifies which woman is meant.^[65] **Parse trees** can be used to show the underlying hierarchy employed to combine the different parts.^[66] Various grammatical devices, like the **gerund** form, also contribute to meaning and are studied by grammatical semantics.^[67]

Formal semantics [[edit](#)]

*Main article: **Formal semantics (natural language)***

Formal semantics uses formal tools from **logic** and **mathematics** to analyze meaning in natural languages.^[f] It aims to develop precise logical formalisms to clarify the relation between expressions and their denotation.^[69] One of its key tasks is to provide frameworks of how language represents the world, for example, using **ontological models** to show how linguistic expressions map to the entities of that model.^[69] A common idea is that words refer to individual objects or groups of objects while sentences relate to events and states. Sentences are mapped to a **truth value** based on whether their description of the world is in correspondence with its ontological model.^[70]

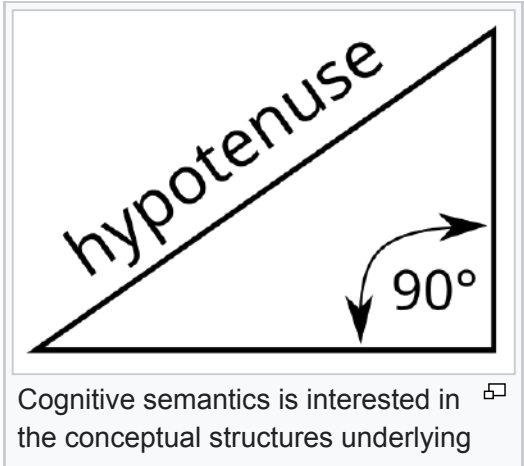
Formal semantics further examines how to use formal mechanisms to represent linguistic phenomena such as **quantification**, **intensionality**, **noun phrases**, **plurals**, mass terms, **tense**, and **modality**.^[71] **Montague semantics** is an early and influential theory in formal semantics that provides a detailed analysis of how the English language can be represented using mathematical logic. It relies on **higher-order logic**, **lambda calculus**, and **type theory** to show how meaning is created through the combination of expressions belonging to different syntactic categories.^[72]

Dynamic semantics is a subfield of formal semantics that focuses on how information grows over time. According to it, "meaning is context change potential": the meaning of a sentence is not given by the information it contains but by the information change it brings about relative to a context.^[73]

Cognitive semantics [[edit](#)]

*Main article: **Cognitive semantics***

Cognitive semantics studies the problem of meaning from a psychological perspective or how the mind of the language user affects meaning. As a subdiscipline of **cognitive linguistics**, it sees language as a wide cognitive ability that is closely related to the conceptual structures used to understand and represent the world.^{[74][g]} Cognitive semanticists do not draw a sharp distinction between linguistic knowledge and knowledge of the world and see them instead as interrelated phenomena.^[76] They study how the interaction between language and human cognition affects the conceptual organization in very general domains like space, time, causation, and action.^[77] The contrast between profile and base is sometimes used to articulate the underlying knowledge structure. The profile of a linguistic expression is the aspect of the knowledge structure that it brings to the foreground while the base is the background that provides the context of this aspect without being at the center of attention.^[78] For example, the profile of the word ***hypotenuse*** is a straight line while the base is a **right-angled triangle** of which the hypotenuse forms a part.^{[79][h]}



Cognitive semantics further compares the conceptual patterns and **linguistic typologies** across languages and considers to what extent the cognitive conceptual structures of humans are universal or relative to their linguistic background.^[81] Another research topic concerns the psychological processes involved in the application of grammar.^[82] Other investigated phenomena include categorization, which is understood as a cognitive heuristic to avoid information overload by regarding different entities in the same way,^[83] and **embodiment**, which concerns how the language user's bodily experience affects the meaning of expressions.^[84]

Frame semantics is an important subfield of cognitive semantics.^[85] Its central idea is that the meaning of terms cannot be understood in isolation from each other but needs to be analyzed on the background of the conceptual structures they depend on. These structures are made explicit in terms of semantic frames. For example, words like bride, groom, and honeymoon evoke in the mind the frame of marriage.^[86]

Others [edit]

Conceptual semantics shares with cognitive semantics the idea of studying linguistic meaning from a psychological perspective by examining how humans conceptualize and experience the world. It holds that meaning is not about the objects to which expressions refer but about the cognitive structure of human concepts that connect thought, perception, and action. Conceptual semantics differs from cognitive semantics by introducing a strict distinction between meaning and syntax and by relying on various formal devices to explore the relation between meaning and cognition.^[87]

Computational semantics examines how the meaning of natural language expressions can be represented and processed on computers.^[88] It often relies on the insights of formal semantics and applies them to problems that can be computationally solved.^[89] Some of its key problems include computing the meaning of complex expressions by analyzing their parts, handling ambiguity, vagueness, and context-dependence, and using the extracted information in **automatic reasoning**.^[90] It forms part of **computational linguistics**, **artificial intelligence**, and **cognitive science**.^[88] Its applications include **machine learning** and **machine translation**.^[91]

Cultural semantics studies the relation between linguistic meaning and culture. It compares conceptual structures in different languages and is interested in how meanings evolve and change because of cultural phenomena associated with **politics**, religion, and **customs**.^[92] For example, address practices encode cultural values and social hierarchies, as in the difference of politeness of expressions like *tu* and *usted* in Spanish or *du* and *Sie* in German in contrast to English, which lacks **these distinctions** and uses the pronoun *you* in either case.^[93] Closely related fields are intercultural semantics, cross-cultural semantics, and comparative semantics.^[94]

Pragmatic semantics studies how the meaning of an expression is shaped by the situation in which it is used. It is based on the idea that communicative meaning is usually context-sensitive and depends on who participates in the exchange, what information they share, and what their **intentions** and background assumptions are. It focuses on communicative actions, of which linguistic expressions only form one part. Some theorists include these topics within the scope of semantics while others consider them part of the distinct discipline of pragmatics.^[95]

Theories of meaning [edit]

Theories of meaning explain what meaning is, what meaning an expression has, and how the relation between expression and meaning is established.^[96]

Referential [edit]

Referential theories state that the meaning of an expression is the entity to which it points.^[97] The meaning of **singular terms** like **names** is the individual to which they refer. For example, the meaning of the name ***George Washington*** is the person with this name.^[98] General terms refer not to a single entity but to the set of objects to which this term applies. In this regard, the meaning of the term *cat* is the set of all cats.^[99] Similarly, verbs usually refer to classes of actions or events and adjectives refer to properties of individuals and events.^[100]

Simple referential theories face problems for meaningful expressions that have no clear referent. Names like ***Pegasus*** and ***Santa Claus*** have meaning even though they do not point to existing entities.^[101] Other difficulties concern cases in which different expressions are about the same entity. For instance, the expressions ***Roger Bannister*** and *the first man to run a four-minute mile* refer to the same person but do not mean exactly the same thing.^[102] This is particularly relevant when talking about beliefs since a person may understand both expressions without knowing that they point to the same entity.^[103] A further problem is given by expressions whose meaning depends on the context, like the deictic terms *here* and *I*.^[104]

To avoid these problems, referential theories often introduce additional devices. Some identify meaning not directly with objects but with functions that point to objects. This additional level has the advantage of taking the context of an expression into account since the same expression may point to one object in one context and to another object in a different context. For example, the reference of the word *here* depends on the location in which it is used.^[105] A closely related approach is **possible world** semantics, which allows expressions to refer not only to entities in the actual world but also to entities in other possible worlds.^[0] According to this view, expressions like *the first man to run a four-minute mile* refer to different persons in different worlds. This view can also be used to analyze sentences that talk about what is possible or what is necessary: possibility is what is true in some possible worlds while necessity is what is true in all possible worlds.^[107]

Ideational [edit]

Ideational theories, also called mentalist theories, are not primarily interested in the reference of expressions and instead explain meaning in terms of the mental states of language users.^[108] One historically influential approach articulated by **John Locke** holds that expressions stand for **ideas** in the speaker's mind. According to this view, the meaning of the word *dog* is the idea that people have of dogs. Language is seen as a medium used to transfer ideas from the speaker to the audience. After having learned the same meaning of signs, the speaker can produce a sign that corresponds to the idea in their mind and the perception of this sign evokes the same idea in the mind of the audience.^[109]

A closely related theory focuses not directly on ideas but on **intentions**.^[110] This view is particularly associated with **Paul Grice**, who observed that people usually communicate to cause some reaction in their audience. He held that the meaning of an expression is given by the intended reaction. This means that communication is not just about decoding what the speaker literally said but requires an understanding of their intention or why they said it.^[111] For example, telling someone looking for petrol that "there is a garage around the corner" has the meaning that petrol can be obtained there because of the speaker's intention to help. This goes beyond the literal meaning, which has no explicit connection to petrol.^[112]

Causal [edit]

Causal theories hold that the meaning of an expression depends on the causes and effects it has.^[113] According to **behaviorist** semantics, also referred to as stimulus-response theory, the meaning of an expression is given by the situation that prompts the speaker to use it and the response it provokes in the audience.^[114] For instance, the meaning of yelling "Fire!" is given by the presence of an uncontrolled fire and attempts to control it or seek safety.^[115] Behaviorist semantics relies on the idea that learning a language consists in adopting behavioral patterns in the form of **stimulus-response pairs**.^[116] One of its key motivations is to avoid private mental entities and define meaning instead in terms of publicly observable language behavior.^[117]

Another causal theory focuses on the meaning of names and holds that a naming event is required to establish the link between name and named entity. This naming event acts as a form of baptism that establishes the first link of a causal chain in which all subsequent uses of the name participate.^[118] According to this view, the name ***Plato*** refers to an ancient Greek philosopher because, at some point, he was originally named this way and people kept using this name to refer to him.^[119] This view was originally formulated by **Saul Kripke** to apply to names only but has been extended to cover other types of speech as well.^[120]

Others [edit]

Truth-conditional semantics analyzes the meaning of sentences in terms of their truth conditions. According to this view, to understand a sentence means to know what the world needs to be like for the sentence to be true.^[121] Truth conditions can themselves be expressed through **possible worlds**. For example, the sentence "**Hillary Clinton** won the **2016 American presidential election**" is false in the actual world but there are some possible worlds in which it is true.^[122] The extension of a sentence can be interpreted as its truth value while its intension is the set of all possible worlds in which it is true.^[123] Truth-conditional semantics is closely related to **verificationist theories**, which introduce the additional idea that there should be some kind of verification procedure to assess whether a sentence is true. They state that the meaning of a sentence consists in the method to verify it or in the circumstances that justify it.^[124] For instance, scientific claims often make predictions, which can be used to confirm or disconfirm them using **observation**.^[125] According to verificationism, sentences that can neither be verified nor falsified are meaningless.^[126]

The **use theory** states that the meaning of an expression is given by the way it is utilized. This view was first introduced by **Ludwig Wittgenstein**, who understood language as a collection of **language games**. The meaning of expressions depends on how they are used inside a game and the same expression may have different meanings in different games.^[127] Some versions of this theory identify meaning directly with patterns of regular use.^[128] Others focus on **social norms** and **conventions** by additionally taking into account whether a certain use is considered appropriate in a given society.^[129]

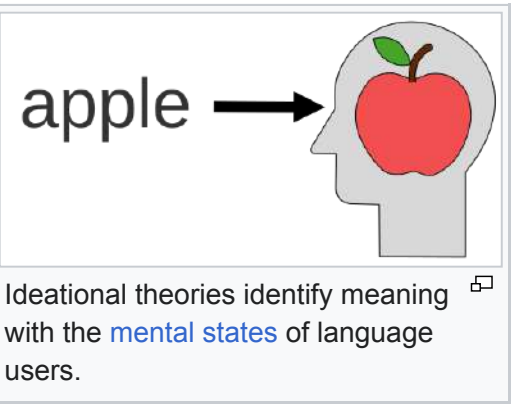
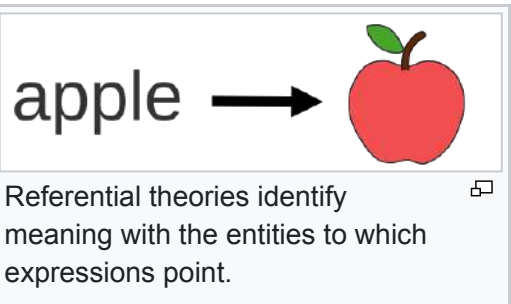
Inferentialist semantics, also called conceptual role semantics, holds that the meaning of an expression is given by the role it plays in the premises and conclusions of good **inferences**.^[130] For example, one can infer from "x is a male sibling" that "x is a brother" and one can infer from "x is a brother" that "x has parents". According to inferentialist semantics, the meaning of the word *brother* is determined by these and all similar inferences that can be drawn.^[131]

History [edit]

Semantics was established as an independent field of inquiry in the 19th century but the study of semantic phenomena began as early as the ancient period as part of philosophy and logic.^{[132][j]} In **ancient Greece**, Plato (427–347 BCE) explored the relation between names and things in his ***dialogue Cratylus***. It considers the positions of naturalism, which holds that things have their name by nature, and conventionalism, which states that names are related to their referents by customs and conventions among language users.^[134] The book ***On Interpretation*** by **Aristotle** (384–322 BCE) introduced various conceptual distinctions that greatly influenced subsequent works in semantics. He developed an early form of the semantic triangle by holding that spoken and written words evoke mental concepts, which refer to external things by resembling them. For him, mental concepts are the same for all humans, unlike the conventional words they associate with those concepts.^[135] The **Stoics** incorporated many of the insights of their predecessors to develop a complex theory of language through the perspective of logic. They discerned different kinds of words by their semantic and syntactic roles, such as the contrast between names, common nouns, and verbs. They also discussed the difference between statements, commands, and prohibitions.^[136]

In **ancient India**, the **orthodox school** of **Nyaya** held that all names refer to real objects. It explored how words lead to an understanding of the thing meant and what consequence this relation has to the creation of knowledge.^[138] Philosophers of the orthodox school of **Mīmāṃsā** discussed the relation between the meanings of individual words and full sentences while considering which one is more basic.^[139] The book ***Vākyapadīya*** by **Bhartrhari** (4th–5th century CE) distinguished between different types of words and considered how they can carry different meanings depending on how they are used.^[140] In **ancient China**, the **Mohists** argued that names play a key role in making distinctions to guide moral behavior.^[141] They inspired the **School of Names**, which explored the relation between names and entities while examining how names are required to identify and judge entities.^[142]

language, which can be articulated through the contrast between profile and base. For instance, the term **hypotenuse** profiles a straight line against the background of a **right-angled triangle**.




 One of **Peter Abelard**'s innovations was his focus on the meaning of full sentences rather than the meaning of individual words.

In the Middle Ages, **Augustine of Hippo** (354–430) developed a general conception of signs as entities that stand for other entities and convey them to the intellect. He was the first to introduce the distinction between natural and linguistic signs as different types belonging to a common genus.^[143] **Boethius** (480–528) wrote a translation of and various comments on Aristotle's book *On Interpretation*, which popularized its main ideas and inspired reflections on semantic phenomena in the **scholastic tradition**.^[144] An innovation in the semantics of **Peter Abelard** (1079–1142) was his interest in propositions or the meaning of sentences in contrast to the focus on the meaning of individual words by many of his predecessors. He further explored the nature of **universals**, which he understood as mere semantic phenomena of **common names** caused by mental abstractions that **do not refer to any entities**.^[145] In the Arabic tradition, **Ibn Faris** (920–1004) identified meaning with the intention of the speaker while **Abu Mansur al-Azhari** (895–980) held that meaning resides directly in speech and needs to be extracted through interpretation.^[146]

An important topic towards the end of the Middle Ages was the distinction between categorematic and **syncategorematic terms**. Categorematic terms have an independent meaning and refer to some part of reality, like *horse* and *Socrates*. Syncategorematic terms lack independent meaning and fulfill other semantic functions, such as modifying or quantifying the meaning of other expressions, like the words *some*, *not*, and *necessarily*.^[147] An early version of the causal theory of meaning was proposed by **Roger Bacon** (c. 1219/20 – c. 1292), who held that things get names similar to how people get names through some kind of initial baptism.^[148] His ideas inspired the tradition of the **speculative grammarians**, who proposed that there are certain universal structures found in all languages. They arrived at this conclusion by drawing an analogy between the modes of signification on the level of language, the modes of understanding on the level of mind, and the modes of being on the level of reality.^[149]

In the early modern period, **Thomas Hobbes** (1588–1679) distinguished between marks, which people use privately to recall their own thoughts, and signs, which are used publicly to communicate their ideas to others.^[150] In their ***Port-Royal Logic***, **Antoine Arnauld** (1612–1694) and **Pierre Nicole** (1625–1695) developed an early precursor of the distinction between intension and extension.^[151] The ***Essay Concerning Human Understanding*** by John Locke (1632–1704) presented an influential version of the ideational theory of meaning, according to which words stand for ideas and help people communicate by transferring ideas from one mind to another.^[152] **Gottfried Wilhelm Leibniz** (1646–1716) understood language as the mirror of thought and tried to conceive the outlines of a **universal formal language** to express scientific and philosophical truths. This attempt inspired theorists **Christian Wolff** (1679–1754), **Georg Bernhard Bilfinger** (1693–1750), and **Johann Heinrich Lambert** (1728–1777) to develop the idea of a general science of sign systems.^[153] **Étienne Bonnot de Condillac** (1715–1780) accepted and further developed Leibniz's idea of the linguistic nature of thought. Against Locke, he held that language is involved in the creation of ideas and is not merely a medium to communicate them.^[154]

In the 19th century, semantics emerged and solidified as an independent field of inquiry. **Christian Karl Reisig** (1792–1829) is sometimes credited as the father of semantics since he clarified its concept and scope while also making various contributions to its key ideas.^[155] **Michel Bréal** (1832–1915) followed him in providing a broad conception of the field, for which he coined the French term *sémantique*.^[156] **John Stuart Mill** (1806–1873) gave great importance to the role of names to refer to things. He distinguished between the connotation and denotation of names and held that propositions are formed by combining names.^[157] **Charles Sanders Peirce** (1839–1914) conceived semiotics as a general theory of signs with several subdisciplines, which were later identified by Charles W. Morris (1901–1979) as syntactics, semantics, and pragmatics. In his pragmatist approach to semantics, Peirce held that the meaning of conceptions consists in the entirety of their practical consequences.^[158] The philosophy of **Gottlob Frege** (1848–1925) contributed to semantics on many different levels. Frege first introduced the distinction between sense and reference, and his development of predicate logic and the principle of compositionality formed the foundation of many subsequent developments in formal semantics.^[159] **Edmund Husserl** (1859–1938) explored meaning from a **phenomenological** perspective by considering the mental acts that endow expressions with meaning. He held that meaning always implies reference to an object and expressions that lack a referent, like *green is or*, are meaningless.^[160]

In the 20th century, **Alfred Tarski** (1901–1983) defined truth in formal languages through his **semantic theory of truth**, which was influential in the development of truth-conditional semantics by **Donald Davidson** (1917–2003).^[161] Tarski's student **Richard Montague** (1930–1971) formulated a complex formal framework of the semantics of the English language, which was responsible for establishing formal semantics as a major area of research.^[162] According to **structural semantics**,^[k] which was inspired by the **structuralist philosophy** of **Ferdinand de Saussure** (1857–1913), language is a complex network of structural relations and the meanings of words are not fixed individually but depend on their position within this network.^[164] The theory of **general semantics** was developed by **Alfred Korzybski** (1879–1950) as an inquiry into how language represents reality and affects human thought.^[165] The contributions of **George Lakoff** (1941–present) and **Ronald Langacker** (1942–present) provided the foundation of cognitive semantics.^[166] **Charles J. Fillmore** (1929–2014) developed frame semantics as a major approach in this area.^[167] The closely related field of conceptual semantics was inaugurated by **Ray Jackendoff** (1945–present).^[168]

In various disciplines [edit]

Logic [edit]

*Main article: **Semantics of logic***

Logicians study correct **reasoning** and often develop **formal languages** to express arguments and assess their correctness.^[169] One part of this process is to provide a semantics for a formal language to precisely define what its terms mean. A semantics of a formal language is a set of rules, usually expressed as a **mathematical function**, that assigns meanings to formal language expressions.^[170] For example, the language of first-order logic uses lowercase letters for **individual constants** and uppercase letters for **predicates**. To express the sentence "Bertie is a dog", the formula ***D**(**b**)* can be used where ***b*** is an individual constant for Bertie and ***D*** is a predicate for dog. Classical model-theoretic semantics assigns meaning to these terms by defining an **interpretation function** that maps individual constants to specific objects and predicates to **sets** of objects or **tuples**. The function maps ***b*** to Bertie and ***D*** to the set of all dogs. This way, it is possible to calculate the truth value of the sentence: it is true if Bertie is a member of the set of dogs and false otherwise.^[171]

Formal logic aims to determine whether arguments are **deductively valid**, that is, whether the premises entail the conclusion.^[172] Entailment can be defined in terms of syntax or in terms of semantics. Syntactic entailment, expressed with the symbol ⊢, relies on **rules of inference**, which can be understood as procedures to transform premises and arrive at a conclusion. These procedures only take the **logical form** of the premises on the level of syntax into account and ignore what meaning they express. Semantic entailment, expressed with the symbol ⊨, looks at the meaning of the premises, in particular, at their truth value. A conclusion follows semantically from a set of premises if the truth of the premises ensures the truth of the conclusion, that is, if any semantic interpretation function that assigns the premises the value *true* also assigns the conclusion the value *true*.^[173]

Computer science [edit]

*Main article: **Semantics (computer science)***

In computer science, the semantics of a **program** is how it behaves when a computer runs it. Semantics contrasts with syntax, which is the particular form in which instructions are expressed. The same behavior can usually be described with different forms of syntax. In **JavaScript**, this is the case for the commands i += 1 and i = i + 1, which are syntactically different expressions to increase the value of the variable *i* by one. This difference is also reflected in different **programming languages** since they rely on different syntax but can usually be employed to create programs with the same behavior on the semantic level.^[174]

Static semantics focuses on semantic aspects that affect the **compilation** of a program. In particular, it is concerned with detecting errors of syntactically correct programs, such as **type errors**, which arise when an operation receives an incompatible **data type**. This is the case, for instance, if a function performing a numerical calculation is given a **string** instead of a number as an argument.^[175] Dynamic semantics focuses on the run time behavior of programs, that is, what happens during the **execution** of instructions.^[176] The main approaches to dynamic semantics are **denotational**, **axiomatic**, and **operational semantics**. Denotational semantics relies on mathematical formalisms to describe the effects of each element of the code. Axiomatic semantics uses deductive logic to analyze which conditions must be in place before and after the execution of a program. Operational semantics interprets the execution of a program as a series of steps, each involving the transition from one **state** to another state.^[177]

Psychology [edit]

*Main article: **Semantics (psychology)***

Psychological semantics examines psychological aspects of meaning. It is concerned with how meaning is represented on a cognitive level and what mental processes are involved in understanding and producing language. It further investigates how meaning interacts with other mental processes, such as the relation between language and perceptual experience.^{[178][l]} Other issues concern how people learn new words and relate them to familiar things and concepts, how they infer the meaning of compound expressions they have never heard before, how they resolve ambiguous expressions, and how semantic illusions lead them to misinterpret sentences.^[180]

One key topic is **semantic memory**, which is a form of **general knowledge** of meaning that includes the knowledge of language, concepts, and facts. It contrasts with **episodic memory**, which records events that a person experienced in their life. The comprehension of language relies on semantic memory and the information it carries about word meanings.^[181] According to a common view, word meanings are stored and processed in relation to their semantic features. The feature comparison model states that sentences like "a robin is a bird" are assessed on a psychological level by comparing the semantic features of the word *robin* with the semantic features of the word *bird*. The assessment process is fast if their semantic features are similar, which is the case if the example is a **prototype** of the general category. For atypical examples, as in the sentence "a penguin is a bird", there is less overlap in the semantic features and the psychological process is significantly slower.^[182]

See also [edit]

- Contronym**
- Semantic technology** – Technology to help machines understand data

References [edit]

Notes [edit]

- ↑ The study of meaning structures found in all languages is sometimes referred to as *universal semantics*.^[5]
- ↑ Semantics usually focuses on **natural languages** but it can also include the study of meaning in **formal languages**, like the language of **first-order logic** and **programming languages**.^[6]
- ↑ *Antonym* is an antonym of *synonym*.^[42]
- ↑ Some linguists use the term *homonym* for both phenomena.^[43]
- ↑ Some authors use the term *compositional semantics* for this type of inquiry.^[60]
- ↑ The term *formal semantics* is sometimes used in a different sense to refer to compositional semantics or to the study of meaning in the formal languages of systems of logic.^[68]
- ↑ Cognitive semantics does not accept the idea of **linguistic relativity** associated with the Sapir–Whorf hypothesis and holds instead that the underlying cognitive processes responsible for conceptual structures are independent of the language one speaks.^[75]


 Bhartṛhari developed and compared various semantic theories of the meaning of words.^[137]

 Michel Bréal coined the French term *sémantique* and conceptualized the scope of this field of inquiry.

- n. [^] Other examples are the word *island*, which profiles a landmass against the background of the surrounding water, and the word *uncle*, which profiles a human adult male against the background of kinship relations.^[80]
- i. [^] A possible world is a complete way of how things could have been.^[106]

Citations [edit]

1.

[^] AHD Staff 2022
Saeed 2009, p. 4
Crimmins 1998, Lead Section

2.

[^] Saeed 2009, pp. 4–5
Crimmins 1998, Lead Section
King 2006, p. 735
Riemer 2010, pp. 2–3

3.

[^] Allan 2009, p. xi

4.

[^] Bezuidenhout 2009, p. 875
Jacobson 2014, p. 4 ↗
Zaefferer 2019, p. 1 ↗

5.

[^] Zaefferer 2019, p. 1 ↗

6.

[^] Lewis 2012, p. 169 ↗
Saeed 2009, pp. 305–306, 315–317
Fernández 2014, p. 14 ↗

7.

[^] ^a ^b Griffiths & Cummins 2023, p. 12

8.

[^] Riemer 2010, p. 2

9.

[^] Crimmins 1998, Lead Section
Saeed 2009, p. 5
Riemer 2010, pp. 4–6

10.

[^] Carston 2011, p. 280 ↗

11.

[^] Williams 1997, p. 457 ↗

12.

[^] Gross 2016, pp. 12–13
Riemer 2016, pp. 30–31

13.

[^] Saeed 2009, pp. 4–5
Jun 2009, p. 463
Jackendoff 2013, p. 72 ↗

14.

[^] Saeed 2009, p. 5
Riemer 2010, pp. 4–6

15.

[^] Griffiths & Cummins 2023, p. 1
Bezuidenhout 2009, p. 875

16.

[^] Griffiths & Cummins 2023, pp. 1–3, 14
Bezuidenhout 2009, p. 875

17.

[^] Griffiths & Cummins 2023, pp. 12–13

18.

[^] Bezuidenhout 2009, p. 875

19.

[^] Malpas 2014, Introduction: Hermeneutics and Philosophy ↗
AHD Staff 2022a
Vámos 2019, p. 3–2 ↗

20.

[^] Anderson 2021, p. 4 ↗
Burgess & Sherman 2014, pp. 1–2 ↗

21.

[^] Riemer 2010, p. 2
Hoad 1993, p. 428

22.

[^] Cunningham 2009, pp. 530–531
Yule 2010, pp. 113–114

23.

[^] Leach & Tartaglia 2018, pp. 274–275
Cunningham 2009, p. 526
Löbner 2013, pp. 1–2
Seachris, § 2a. The Meanings of 'Meaning'

24.

[^] Riemer 2010, pp. 21–22
Griffiths & Cummins 2023, pp. 5–6
Löbner 2013, pp. 1–6, 18–21

25.

[^] Tondl 2012, p. 111

26.

[^] Olkowski & Pirovolakis 2019, pp. 65–66 ↗

27.

[^] Riemer 2010, pp. 21–22
Griffiths & Cummins 2023, pp. 5–6
Löbner 2013, pp. 1–6
Saeed 2009, pp. 12–13

28.

[^] Yule 2010, p. 113
Griffiths & Cummins 2023, pp. 5–6

29.

[^] Zalta 2022, § 1. Frege's Life and Influences, § 3. Frege's Philosophy of Language.

30.

[^] Griffiths & Cummins 2023, pp. 7–9
Cunningham 2009, p. 526
Saeed 2009, p. 46

31.

[^] Cunningham 2009, p. 527
Griffiths & Cummins 2023, pp. 7–9

32.

[^] Cunningham 2009, p. 526
Griffiths & Cummins 2023, pp. 7–9

33.

[^] Marti 1998, Lead Section
Riemer 2010, pp. 27–28

34.

[^] Riemer 2010, pp. 25–28
Griffiths & Cummins 2023, pp. 7–9

35.

[^] Cunningham 2009, p. 531

36.

[^] Marti 1998, Lead Section

37.

[^] ^a ^b ^c Szabó 2020, Lead Section
Pelletier 1994, pp. 11–12
Krifka 2001, p. 152 ↗

38.

[^] Löbner 2013, pp. 7–8, 10–12

39.

[^] ^a ^b Gregory 2016, pp. 9–10
Blackburn 2008, Truth Conditions ↗
Kearns 2011, pp. 8–10

40.

[^] ^a ^b Palmer 1976, pp. 25–26
Noth 1990, pp. 89–90 ↗
Dirven & Verspoor 2004, pp. 28–29 ↗
Riemer 2010, pp. 13–16

41.

[^] Yule 2010, pp. 115–116
Saeed 2009, pp. 152–155

42.

[^] Heffer 2014, p. 42 ↗

43.

[^] Saeed 2009, p. 63

44.

[^] ^a ^b Yule 2010, pp. 116–120
Saeed 2009, pp. 63–70

45.

[^] Edmonds 2009, pp. 223–226
Murphy & Koskela 2010, p. 57 ↗

46.

[^] Yule 2010, pp. 113–115

47.

[^] Saeed 2009, p. 63
Reif & Polzenhagen 2023, pp. 129–130 ↗

48.

[^] Meulen 2008, Philosophy of Language and Linguistics ↗
Kuche & Rowland 2023, pp. 13–14 ↗

49.

[^] Cornish 1999, pp. 18–19 ↗
Jiang 2016, § Summary

50.

[^] Blackburn 2008c
Partee 1997, pp. 833–834 ↗

51.

[^] Riemer 2010, pp. 22–23
Gamut 1991, pp. 142–143 ↗
Dummett 1981, p. 106 ↗

j.

[^] The history of semantics is different from historical semantics, which studies how the meanings of words change through time.^[133]

k.

[^] Some theorists use the term *structural semantics* in a different sense to refer to phrasal semantics.^[163]

52.

[^] Geeraerts 2017, Lead Section
Taylor 2017, pp. 246–247 ↗
Pustejovsky 2006, pp. 98–100 ↗
Pustejovsky 2009, p. 476
Márquez 2011, p. 146 ↗

53.

[^] Geeraerts 2017, Lead Section, § 1. The Descriptive Scope of Lexical Semantics
Pustejovsky 2009, p. 476
Márquez 2011, pp. 146–147 ↗

54.

[^] Geeraerts 2017, Lead Section, § 1.3 Lexical Fields and Componential Analysis
Yule 2010, pp. 113–115

55.

[^] Pustejovsky 2009, p. 479

56.

[^] Pustejovsky 2006, pp. 98–100 ↗
Geeraerts 2017, § 1.1 Polysemy and Vagueness

57.

[^] Geeraerts 2017, § 1. The Descriptive Scope of Lexical Semantics
Noth 1990, p. 106
Taylor 2017, pp. 246–247 ↗

58.

[^] L'Homme 2020, pp. 67–69 ↗
Trips 2009, p. 236 ↗
Andreou 2015, Abstract

59.

[^] Fasold & Connor-Linton 2006, pp. 141, 156 ↗
Jackendoff 2002, p. 378
Park-Johnson & Shin 2020, pp. 103–104
Riemer 2010, p. 21
Bieswanger & Becker 2017, p. 128 ↗
Jacobson 2014, p. 5

60.

[^] Fasold & Connor-Linton 2006, p. 156 ↗
Bagha 2011, pp. 1414–1415

61.

[^] ^a ^b Fasold & Connor-Linton 2006, pp. 141–143 ↗

62.

[^] Fasold & Connor-Linton 2006, p. 143 ↗
Park-Johnson & Shin 2020, pp. 103–104

63.

[^] Jackendoff 2002, pp. 378–380

64.

[^] Jackendoff 2002, pp. 382–383

65.

[^] Jackendoff 2002, pp. 384–385

66.

[^] Valin 2008, p. 466 ↗
Berwick & Stabler 2019, p. 110 ↗

67.

[^] Wierzbicka 1988, p. 3 ↗

68.

[^] Bohnemeyer 2021, p. 24 ↗
Pollock 2017, p. 172 ↗

69.

[^] ^a ^b Geeraerts 2010, pp. 118–119 ↗
Moeschler 2007, pp. 31–33 ↗
Portner & Partee 2008, pp. 1–2 ↗

70.

[^] Moeschler 2007, pp. 31–33 ↗

71.

[^] Portner & Partee 2008, pp. 3, 8–10, 35, 127, 324 ↗

72.

[^] Portner & Partee 2008, pp. 3–4 ↗
Janssen & Zimmermann 2021, Lead Section, § 1. Introduction, § 2.3 Logic and Translation

73.

[^] Groenendijk & Stokhof 2009, pp. 272–273
Nouwen et al. 2022, Lead Section

74.

[^] Li 2021
Taylor 2009, pp. 73–74
Croft & Cruse 2004, pp. 1–3 ↗

75.

[^] Kortmann 2020, p. 165 ↗.

76.

[^] Taylor 2009, pp. 73–74

77.

[^] Li 2021

78.

[^] Taylor 2009, pp. 74–75
Enfield 2002, p. 152 ↗

79.

[^] Taylor 2009, pp. 74–75
Taylor 2013, pp. 38–40 ↗

80.

[^] Taylor 2009, pp. 74–75

81.

[^] Taylor 2009, p. 85
Li 2021

82.

[^] Li 2021
Taylor 2009, pp. 83–84

83.

[^] Taylor 2009, pp. 76–77

84.

[^] Taylor 2009, p. 82

85.

[^] Mushayabasa 2014, p. 21 ↗
Shead 2011, pp. 34–35 ↗

86.

[^] Gawron 2011, pp. 664–665, 669
Fillmore 2009, pp. 330–332

87.

[^] Riemer 2010, pp. 261–263
Jackendoff 2011, p. 688

88.

[^] ^a ^b Geeraerts 2010, p. 118 ↗
Bunt & Muskens 1999, pp. 1–2 ↗

89.

[^] Bunt & Muskens 1999, pp. 1–2 ↗

90.

[^] Geeraerts 2010, p. 118 ↗
Bunt & Muskens 1999, pp. 1–2 ↗
Erk 2018, Summary

91.

[^] Erk 2018, Summary
Geeraerts 2010, p. 118 ↗

92.

[^] Zhao 2023, Preface ↗

93.

[^] Farese 2018, pp. 1–3 ↗

94.

[^] Peeters 2006, p. 25 ↗

95.

[^] Márquez 2011, p. 149 ↗
Bublitz & Norrick 2011, pp. 215–216 ↗

96.

[^] Speaks 2021, Lead Section
Glock 2012, pp. 51–52
Holm & Karlgren 1995, pp. 20–21
Bagha 2011, pp. 1414–1415

97.

[^] Glock 2012, p. 51
Holm & Karlgren 1995, pp. 21–22

98.

[^] Speaks 2021, § 2.1.1 The Theory of Reference
Holm & Karlgren 1995, pp. 21–22
Davis 2005, pp. 209–210
Gibbs 1994, pp. 29–30 ↗

99.

[^] Davis 2005, pp. 209–210

100.

[^] Gibbs 1994, pp. 29–30 ↗

101.

[^] Davis 2005, p. 211

102.

[^] Holm & Karlgren 1995, pp. 21–22
Gibbs 1994, pp. 29–30 ↗

103.

[^] Speaks 2021, § 2.1.2 Theories of Reference Vs. Semantic Theories

104.

[^] Speaks 2021, § 2.1.4 Character and Content, Context and Circumstance

l.

[^] Some theorists use the term *psychosemantics* to refer to this discipline while others understand the term in a different sense.^[179]

105.

[^] Davis 2005, pp. 209–210
Holm & Karlgren 1995, pp. 21–22
Speaks 2021, § 2.1.4 Character and Content, Context and Circumstance

106.

[^] Berto & Jago 2023, Lead Section
Pavel 1986, p. 50 ↗

107.

[^] Speaks 2021, § 2.1.5 Possible Worlds Semantics
Rooij 2012, pp. 198–199 ↗
Davis 2005, pp. 209–210

108.

[^] Chapman & Routledge 2009
Speaks 2021, § 3.1 Mentalist Theories

109.

[^] Chapman & Routledge 2009
Holm & Karlgren 1995, p. 22
Appiah & Gutmann 1998, p. 34 ↗
Pearce 2022, pp. 194–195 ↗

110.

[^] Chapman & Routledge 2009
Pearce 2022, pp. 194–195 ↗
Speaks 2021, § 3.1.1 The Gricean Program

111.

[^] Chapman & Routledge 2009
Glock 2012, p. 52
Speaks 2021, § 3.1.1 The Gricean Program
Feng 2010, pp. 11–12 ↗

112.

[^] Feng 2010, p. 19 ↗

113.

[^] Glock 2012, pp. 51–52
Blackburn 2008a
Speaks 2021, § 3.2.1 Causal Origin

114.

[^] Holm & Karlgren 1995, p. 23
Lyons 1996, pp. 120, 123–125
Lepore 2009, p. 1026

115.

[^] Duignan 2023, § Behaviourist Semantics
Lepore 2009, p. 1026

116.

[^] Lyons 1996, pp. 123–125

117.

[^] Lyons 1996, pp. 120–121

118.

[^] Blackburn 2008a
Speaks 2021, § 3.2.1 Causal Origin

119.

[^] Blackburn 2008a

120.

[^] Speaks 2021, § 3.2.1 Causal Origin

121.

[^] Dummett 2008, pp. 45–46
Kearns 2011, pp. 8–11

122.

[^] Berto & Jago 2023, § 1. Reasons for Introducing Impossible Worlds

123.

[^] Kearns 2011, pp. 8–11

124.

[^] Glock 2012, p. 51
Morris & Preti 2023, pp. 369–370 ↗
Boyd, Gasper & Trout 1991, p. 5 ↗
Lepore 2009, p. 1027

125.

[^] Boyd, Gasper & Trout 1991, p. 5 ↗

126.

[^] Morris & Preti 2023, pp. 369–370 ↗
Boyd, Gasper & Trout 1991, p. 5 ↗

127.

[^] Holm & Karlgren 1995, pp. 23–24
Strauven 2018, p. 78 ↗

128.

[^] Speaks 2021, § 3.2.4 Regularities in Use

129.

[^] Speaks 2021, § 3.2.5 Social Norms

130.

[^] Speaks 2021, § 2.2.3 Inferentialist Semantics
Whiting, Lead Section, § 1a. A Theory of Linguistic Meaning
Hess 2022, § Abstract, § 1 Introduction

131.

[^] Whiting, § 1a. A Theory of Linguistic Meaning

132.

[^] Meier-Oeser 2019, p. 182
Nerlich 1992, p. 2 ↗

133.

[^] Geeraerts 2010, p. 1 ↗
Kay 2015, pp. 1–2 ↗

134.

[^] Meier-Oeser 2019, pp. 184–185
Allan 2015, p. 48
Kretzmann 2006, pp. 752–753

135.

[^] Meier-Oeser 2019, pp. 186–187
Kretzmann 2006, pp. 755–756

136.

[^] Meier-Oeser 2019, pp. 189–191
Kretzmann 2006, pp. 757–758
Durand 2018, p. ii

137.

[^] Bekkum et al. 1997, pp. 110–112 ↗

138.

[^] Bekkum et al. 1997, p. 102 ↗
Chakrabarti 1997, p. 215 ↗

139.

[^] Bekkum et al. 1997, pp. 75–76 ↗

140.

[^] Aklujkar 1970, p. 13
Staal 1966, pp. 304–307
Bekkum et al. 1997, pp. 110–112 ↗
Cardona 2019, p. 303 ↗

141.

[^] Fraser 2023, pp. 204–205 ↗
Indraccolo 2020, p. 174 ↗

142.

[^] Sun 2014, p. 23 ↗
Fraser 2020, Lead Section, § 2.1 Same and Different

143.

[^] Meier-Oeser 2019, p. 192
Kretzmann 2006, pp. 759–761
Benin 2012, p. 94 ↗

144.

[^] Meier-Oeser 2019, pp. 193–195
Kretzmann 2006, pp. 761–762

145.

[^] Meier-Oeser 2019, pp. 197–198
Kretzmann 2006, pp. 763–764
Provenzola 2013, p. 22 ↗

146.

[^] Bekkum et al. 1997, p. 229 ↗
Leaman 2015, p. 180 ↗

147.

[^] Meier-Oeser 2019, pp. 198–199
Kretzmann 2006, pp. 769–770
Blackburn 2008b

148.

[^] Meier-Oeser 2019, pp. 200–201
Kretzmann 2006, pp. 769–770

149.

[^] Meier-Oeser 2019, pp. 201–202
Kretzmann 2006, pp. 770–771

150.

[^] Meier-Oeser 2019, pp. 205–206
Kretzmann 2006, pp. 773–774
Martin 1953, p. 210

151.

[^] Meier-Oeser 2019, pp. 206–207
Kretzmann 2006, p. 777

152.

[^] Meier-Oeser 2019, pp. 207–208
Kretzmann 2006, pp. 777–778
Chapman & Routledge 2009

153.

[^] Meier-Oeser 2019, pp. 209–210
Kretzmann 2006, pp. 779–780

154. ↑ Meier-Oeser 2019, pp. 212–213

Kretzmann 2006, pp. 784–785

155. ↑ Nerlich 2019, pp. 218, 221–223

156. ↑ Nerlich 2019, pp. 230–231

Allan 2015, p. 51

157. ↑ Kretzmann 2006, pp. 795–796

158. ↑ Burch & Parker 2024, § 4. Pragmatism, Pragmaticism, and the Scientific Method

Kretzmann 2006, pp. 797–799

159. ↑ Kretzmann 2006, pp. 787–789

Pelletier 1994, Abstract

160. ↑ Kretzmann 2006, pp. 802–803

161. ↑ Harris 2017, pp. 150, 162

Gupta 2011, p. 236↗

Lepore & Ludwig 2009, pp. 1–2↗

162. ↑ Harris 2017, pp. 150, 162

Chatzikyriakidis & Luo 2021, p. 6↗

163. ↑ Rowe & Levine 2015, p. 151↗

164. ↑ Murphy 2009, p. 27

Noth 1990, p. 61

165. ↑ Hörmann 2013, p. 298↗

Steinfatt 2009, pp. 438–440↗

Sources [edit]

- AHD Staff (2022). "Semantics"↗. *American Heritage Dictionary*. Harper Collins. Archived↗ from the original on 31 January 2024. Retrieved 31 January 2024.
- AHD Staff (2022a). "Hermeneutics"↗. *American Heritage Dictionary*. Harper Collins. Archived↗ from the original on 23 February 2024. Retrieved 31 January 2024.
- Aklujkar, Ashok (1970). "Ancient Indian Semantics". *Annals of the Bhandarkar Oriental Research Institute*. **51** (1/4): 11–29. ISSN 0378-1143. JSTOR 41688671↗.
- Allan, Keith (2009). "Introduction"↗. In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Allan, Keith (2015). "3. A History of Semantics"↗. In Riemer, Nick (ed.). *The Routledge Handbook of Semantics*. Routledge. ISBN 978-1-317-41245-8. Retrieved 2024-02-23.
- Anderson, Derek Egan (2021). *Metasemantics and Intersectionality in the Misinformation Age: Truth in Political Struggle*↗. Springer Nature. ISBN 978-3-030-73339-1. Retrieved 2024-02-23.
- Andreou, Marios (2015). "Lexical Negation in Lexical Semantics: The Prefixes in- and dis-". *Morphology*. **25** (4): 391–410. doi:10.1007/s11525-015-9266-z↗.
- Appiah, Kwame Anthony; Gutmann, Amy (1998). *Color Conscious: The Political Morality of Race*↗. Princeton University Press. ISBN 978-1-400-82209-6. Retrieved 2024-02-18.
- Bagha, Karim Nazari (2011). "A Short Introduction to Semantics". *Journal of Language Teaching and Research*. **2** (6). doi:10.4304/jltr.2.6.1411-1419↗.
- Bekkum, Wout Jac van; Houben, Jan; Sluiter, Ineke; Versteegh, Kees (1997). *The Emergence of Semantics in Four Linguistic Traditions: Hebrew, Sanskrit, Greek, Arabic*↗. John Benjamins. ISBN 978-9-027-29881-2. Retrieved 2024-02-23.
- Benin, Stephen D. (2012). *The Footprints of God: Divine Accommodation in Jewish and Christian Thought*↗. State University of New York Press. ISBN 978-0-791-49628-2. Retrieved 2024-02-23.
- Berto, Francesco; Jago, Mark (2023). "Impossible Worlds"↗. *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived↗ from the original on 10 February 2021. Retrieved 17 February 2024.
- Berwick, Robert C.; Stabler, Edward P. (2019). *Minimalist Parsing*↗. Oxford University Press. ISBN 978-0-198-79508-7. Retrieved 2024-02-15.
- Bezuidenhout, A. (2009). "Semantics–Pragmatics Boundary"↗. In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Bieswanger, Markus; Becker, Annette (2017). *Introduction to English Linguistics*↗. UTB. ISBN 978-3-825-24528-3. Retrieved 2024-02-15.
- Blackburn, Simon (2008). "Truth Conditions". *The Oxford Dictionary of Philosophy*↗. Oxford University Press. ISBN 978-0-199-54143-0. Archived↗ from the original on 2024-02-08. Retrieved 2024-02-09.
- Blackburn, Simon (2008a). "Causal Theory of Meaning". *The Oxford Dictionary of Philosophy*↗. Oxford University Press. ISBN 978-0-199-54143-0. Archived↗ from the original on 2024-02-17. Retrieved 2024-02-18.
- Blackburn, Simon (2008b). "Syncategorematic". *The Oxford Dictionary of Philosophy*↗. Oxford University Press. ISBN 978-0-199-54143-0. Archived↗ from the original on 2024-02-23. Retrieved 2024-02-23.
- Blackburn, Simon (2008c). "Referentially Opaque/Transparent". *The Oxford Dictionary of Philosophy*↗. Oxford University Press. ISBN 978-0-199-54143-0. Archived↗ from the original on 2024-02-24. Retrieved 2024-02-24.
- Bohnenmeyer, Jürgen (2021). *Ten Lectures on Field Semantics and Semantic Typology*↗. Brill. ISBN 978-9-004-36262-8. Retrieved 2024-02-15.
- Boyd, Richard; Gasper, Philip; Trout, J. D. (1991). *The Philosophy of Science*↗. MIT Press. ISBN 978-0-262-52156-7. Retrieved 2024-02-18.
- Bublitz, Wolfram; Norrick, Neal R. (2011). *Foundations of Pragmatics*↗. Walter de Gruyter. ISBN 978-3-110-21426-0. Retrieved 2024-02-15.
- Bunt, Harry; Muskens, Reinhard (1999). "Computational Semantics"↗. *Computing Meaning: Volume 1*. Springer Netherlands. doi:10.1007/978-94-011-4231-1_1↗. ISBN 978-9-401-14231-1. Retrieved 2024-02-15.
- Burch, Robert; Parker, Kelly A. (2024). "Charles Sanders Peirce"↗. *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived↗ from the original on 7 January 2020. Retrieved 22 February 2024.
- Burgess, Alexis; Sherman, Brett (2014). "Introduction: A Plea for the Metaphysics of Meaning". In Burgess, Alexis; Sherman, Brett (eds.). *Metasemantics: New Essays on the Foundations of Meaning*↗. Oxford University Press. ISBN 978-0-191-64835-9. Retrieved 2024-02-23.
- Cardona, Georgio R. (2019). *Panini: A Survey of Research*↗. Walter de Gruyter. ISBN 978-3-110-80010-4. Retrieved 2024-02-23.
- Carston, Robyn (2011). "Truth-conditional Semantics"↗. In Sbisà, Marina; Östman, Jan-Ola; Verschueren, Jef (eds.). *Philosophical Perspectives for Pragmatics*. John Benjamins. ISBN 978-9-027-20787-6. Retrieved 2024-02-10.

166. ↑ Taylor 2009, p. 7↗

Fischer 2013, p. 7↗

167. ↑ Croft & Cruse 2004, p. 8↗

168. ↑ Östman & Fried 2005, pp. 191–192↗

169. ↑ Riemer 2010, pp. 173–174

Jaakko & Sandu 2006, pp. 13–14

Shapiro & Kouri Kissel 2024, Lead Section, § 2. Language

170. ↑ Shapiro & Kouri Kissel 2024, Lead Section, § 4. Semantics

Jansana 2022, § 5. Algebraic Semantics

Jaakko & Sandu 2006, pp. 17–18

171. ↑ Grimm 2009, pp. 116–117↗

Shapiro & Kouri Kissel 2024, Lead Section, § 4. Semantics

Magnus et al. 2021, pp. 193–195

172. ↑ Riemer 2010, pp. 173–174

Jaakko & Sandu 2006, pp. 13–14

Shapiro & Kouri Kissel 2024, Lead Section

Gregory 2017, p. 82↗

173. ↑ Forster 2003, pp. 74–75↗

Johnstone 1987, p. 23↗

Shapiro & Kouri Kissel 2024, Lead Section, § 4. Semantics

Jaakko & Sandu 2006, pp. 17–20

174. ↑ Fernández 2014, pp. 10, 14–15↗

Dale, Weems & Headington 2003, pp. 42, 44↗

O'Regan 2020, p. 193↗

- Chakrabarti, A. (1997). *Denying Existence: The Logic, Epistemology and Pragmatics of Negative Existentials and Fictional Discourse*↗. Springer. ISBN 978-0-792-34388-2. Retrieved 2024-02-23.
- Chapman, Siobhan; Routledge, Christopher (2009). "Ideational Theories". *Key Ideas in Linguistics and the Philosophy of Language*↗. Edinburgh University Press. pp. 84–85. doi:10.1515/9780748631421-033↗. ISBN 978-0-748-63142-1. Retrieved 2024-02-18.
- Chatzikyriakidis, Stergios; Luo, Zhaohui (2021). *Formal Semantics in Modern Type Theories*↗. John Wiley & Sons. ISBN 978-1-786-30128-4. Retrieved 2024-02-23.
- Cohen, Jonathan (2009). *The Red and the Real: An Essay on Color Ontology*↗. Oxford University Press. ISBN 978-0-191-60960-2. Retrieved 2024-02-19.
- Cornish, Francis (1999). *Anaphora, Discourse, and Understanding: Evidence from English and French*↗. Clarendon Press. ISBN 978-0-198-70028-9. Retrieved 2024-02-24.
- Crimmins, Mark (1998). "Semantics"↗. *Routledge Encyclopedia of Philosophy*. Routledge. doi:10.4324/9780415249126-U036-1↗. ISBN 978-0-415-25069-6. Retrieved 2024-02-18.
- Croft, William; Cruse, D. Alan (2004). *Cognitive Linguistics*↗. Cambridge University Press. ISBN 978-0-521-66770-8. Retrieved 2024-02-23.
- Cunningham, D. J. (2009). "Meaning, Sense, and Reference"↗. In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Dale, Neil B.; Weems, Chip; Headington, Mark R. (2003). *Programming and Problem Solving with Java*↗. Jones & Bartlett. ISBN 978-0-763-70490-2. Retrieved 2024-02-18.
- Davis, Wayne A. (2005). *Nondescriptive Meaning and Reference: An Ideational Semantics*↗. Oxford University Press. ISBN 978-0-191-60309-9. Archived↗ from the original on 2024-02-16. Retrieved 2024-02-18.
- Dirven, René; Verspoor, Marjolijn (2004). *Cognitive Exploration of Language and Linguistics*↗ (2nd ed.). John Benjamins. ISBN 978-9-027-29541-5. Retrieved 2024-02-09.
- Duignan, Brian (2023). "Semantics"↗. *Encyclopædia Britannica*. Archived↗ from the original on 5 December 2023. Retrieved 17 February 2024.
- Dummett, Michael (1981). *Frege: Philosophy of Language*↗. Harvard University Press. ISBN 978-0-674-31931-8. Retrieved 2024-02-09.
- Dummett, Michael (2008). *Thought and Reality*. Clarendon. ISBN 978-0-199-20727-5. Retrieved 2024-02-15.
- Durand, Marion Olivia (2018). *Language and Reality: Stoic Semantics Reconstructed*↗ (Thesis). University of Toronto.
- Edmonds, P. (2009). "Disambiguation"↗. In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Enfield, N. J. (2002). *Ethnosyntax: Explorations in Grammar and Culture: Explorations in Grammar and Culture*↗. Oxford University Press. ISBN 978-0-191-58179-3. Retrieved 2024-02-15.
- Erk, Katrin (2018). "Computational Semantics"↗. *Oxford Research Encyclopedia of Linguistics*. Oxford University Press. ISBN 978-0-199-38465-5. Archived↗ from the original on 2024-02-13. Retrieved 2024-02-15.
- Farese, Gian Marco (2018). *The Cultural Semantics of Address Practices: A Contrastive Study Between English and Italian*↗. Rowman & Littlefield. ISBN 978-1-498-57928-5. Retrieved 2024-02-15.
- Fasold, Ralph; Connor-Linton, Jeffrey (2006). *An Introduction to Language and Linguistics*↗. Cambridge University Press. ISBN 978-1-107-71766-4. Retrieved 2024-02-15.
- Feng, Guangwu (2010). *A Theory of Conventional Implicature and Pragmatic Markers in Chinese*↗. Brill. ISBN 978-1-849-50934-3. Retrieved 2024-02-18.
- Fernández, Maribel (2014). *Programming Languages and Operational Semantics: A Concise Overview*↗. Springer. ISBN 978-1-447-16368-8. Retrieved 2024-02-04.
- Fillmore, C. J. (2009). "Frame Semantics". In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*↗. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Fischer, Kerstin (2013). *From Cognitive Semantics to Lexical Pragmatics: The Functional Polysemy of Discourse Particles*↗. Walter de Gruyter. ISBN 978-3-110-82864-1. Retrieved 2024-02-23.
- Forster, Thomas (2003). *Logic, Induction and Sets*↗. Cambridge University Press. ISBN 978-0-521-53361-4. Retrieved 2024-02-19.
- Fraser, Chris (2020). "School of Names"↗. *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University.
- Fraser, Chris (2023). *Late Classical Chinese Thought*↗. Oxford University Press. ISBN 978-0-192-59168-5. Retrieved 2024-02-15.
- Fritzson, Peter (2010). *Principles of Object-Oriented Modeling and Simulation with Modelica 2.1*↗. John Wiley & Sons. ISBN 978-0-470-93761-7. Retrieved 2024-02-19.
- Gamut, L. T. F. (1991). *Logic, Language, and Meaning, Volume 1: Introduction to Logic*↗. University of Chicago Press. ISBN 978-0-226-28084-4. Retrieved 2024-02-09.
- Gawron, Jean-Mark (2011). "29. Frame Semantics". *Semantics*↗. De Gruyter Mouton. ISBN 978-3-110-22661-4. Retrieved 2024-02-15.

175. ↑ Fernández 2014, pp. 14–15↗

Fritzson 2010, p. 703↗

Mosses 2003, p. 167↗

176. ↑ Fernández 2014, pp. 15–16↗

Fritzson 2010, p. 703↗

Mosses 2003, p. 167↗

177. ↑ Fernández 2014, p. 16↗

O'Regan 2020, pp. 193–194↗

178. ↑ Smith, Rips & Shoben 1975, pp. 1–3↗

Sanford 2009, pp. 792–793, 796

179. ↑ Halpern & Voïskunskii 1997, p. 21↗

Cohen 2009, p. 59↗

180. ↑ Sanford 2009, pp. 793–797

181. ↑ Smith, Rips & Shoben 1975, pp. 3–4, 42↗

Hampton 2015, p. 125↗

Tulving 2001, p. 278↗

182. ↑ Sanford 2009, p. 792

Smith, Rips & Shoben 1975, pp. 3–4, 42↗

Hampton 2015, pp. 125–128↗

Shi 2017, pp. 85–86↗

- Geeraerts, Dirk (2010). *Theories of Lexical Semantics*↗. Oxford University Press. ISBN 978-0-198-70030-2. Retrieved 2024-02-15.
- Geeraerts, Dirk (2017). "Lexical Semantics"↗. *Oxford Research Encyclopedia of Linguistics*. Oxford University Press. ISBN 978-0-199-38465-5. Archived↗ from the original on 2024-02-15. Retrieved 2024-02-15.
- Gibbs, Raymond W. (1994). *The Poetics of Mind: Figurative Thought, Language, and Understanding*↗. Cambridge University Press. ISBN 978-0-521-42992-4. Retrieved 2024-02-18.
- Glock, Hans-Johann (2012). "What Is a Theory of Meaning? Just When You Thought Conceptual Analysis Was Dead...". *Cahiers Ferdinand de Saussure* (65): 51–79. ISSN 0068-516X↗. JSTOR 24324915↗.
- Gregory, Howard (2016). *Semantics*. Routledge. ISBN 978-0-415-21610-4. Retrieved 2024-02-18.
- Gregory, Paul A. (2017). *Formal Logic*↗. Broadview Press. ISBN 978-1-77048-594-5. Retrieved 2024-02-19.
- Griffiths, Patrick; Cummins, Chris (2023). *An Introduction to English Semantics and Pragmatics* (3rd ed.). Edinburgh University Press. ISBN 978-1-399-50460-7. Retrieved 2024-02-18.
- Grimm, Stephan (2009). "Knowledge Representation and Ontologies"↗. In Gaber, Mohamed Medhat (ed.). *Scientific Data Mining and Knowledge Discovery: Principles and Foundations*. Springer. ISBN 978-3-642-02788-8. Retrieved 2024-02-19.
- Groenendijk, J.; Stokhof, M. (2009). "Dynamic Semantics"↗. In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Gross, Steven (2016). "(Descriptive) Externalism in Semantics". In Riemer, Nick (ed.). *The Routledge Handbook of Semantics*. Routledge. ISBN 978-0-415-66173-7. Retrieved 2024-02-18.
- Gupta, Anil (2011). "An Argument Against Tarski's Convention T"↗. In Schantz, Richard (ed.). *What Is Truth?*. Walter de Gruyter. ISBN 978-3-110-88666-5. Retrieved 2024-02-23.
- Halpern, Diane F.; Voïskunskii, Aleksandr (1997). *States of Mind: American and Post-Soviet Perspectives on Contemporary Issues in Psychology*↗. Oxford University Press. ISBN 978-0-19-510351-9. Retrieved 2024-02-19.
- Hampton, James A. (2015). "7. Categories, Prototypes, and Exemplars"↗. In Riemer, Nick (ed.). *The Routledge Handbook of Semantics*. Routledge. ISBN 978-1-317-41245-8. Retrieved 2024-02-19.
- Harris, Daniel W. (2017). "The History and Prehistory of Natural-Language Semantics"↗. *Innovations in the History of Analytical Philosophy*. Palgrave Macmillan. pp. 149–194. doi:10.1057/978-1-137-40808-2_6↗. ISBN 978-1-137-40808-2. Retrieved 2024-02-23.
- Heffer, Simon (2014). *Simply English: An A-Z of Avoidable Errors*↗. Random House. ISBN 978-1-446-47380-1. Retrieved 2024-02-19.
- Hess, Leopold (2022). "Inferentialist Semantics for Lexicalized Social Meanings"↗. *Synthese*. **200** (5). doi:10.1007/s11229-022-03817-5↗.
- Hoad, T. F. (1993). *The Concise Oxford Dictionary of English Etymology*. Oxford University Press. ISBN 978-0-192-83098-2. Retrieved 2024-02-18.
- Holm, P.; Karlgren, K. (1995). "Theories of Meaning and Different Perspectives on Information Systems"↗. *Information System Concepts: Towards a Consolidation of Views*. Springer US. doi:10.1007/978-0-387-34870-4_3↗. ISBN 978-0-387-34870-4. Archived↗ from the original on 2024-02-16. Retrieved 2024-02-18.
- Hörmann, Hans (2013). *Psycholinguistics: An Introduction to Research and Theory*↗. Springer Science & Business Media. ISBN 978-1-461-26211-4. Retrieved 2024-02-24.
- Indraccolo, Lisa (2020). "Argumentation (Bian 辯)". In Fung, Yiu-ming (ed.). *Dao Companion to Chinese Philosophy of Logic*↗. Dao Companions to Chinese Philosophy. Vol. 12. Springer Nature. ISBN 978-3-030-29033-7. Retrieved 2024-02-19.
- Jaakko, Hintikka; Sandu, Gabriel (2006). "What Is Logic?". *Philosophy of Logic*↗. Elsevier. ISBN 978-0-080-46663-7. Retrieved 2024-02-19.
- Jackendoff, Ray (2013). "Constructions in the Parallel Architecture"↗. In Hoffmann, Thomas; Trousdale, Graeme (eds.). *The Oxford Handbook of Construction Grammar*. Oxford University Press. ISBN 978-0-199-37663-6. Retrieved 2024-02-04.
- Jackendoff, Ray (2011). "30. Conceptual Semantics". *Semantics*. De Gruyter Mouton. doi:10.1515/9783110226614.688↗. ISBN 978-3-110-22661-4. Retrieved 2024-02-15.
- Jackendoff, Ray (2002). *Foundations of Language: Brain, Meaning, Grammar, Evolution*. Oxford University Press. ISBN 978-0-198-27012-6. Retrieved 2024-02-18.
- Jacobson, Pauline I. (2014). *Compositional Semantics: An Introduction to the Syntax/Semantics Interface*↗. Oxford University Press. ISBN 978-0-199-67714-6. Retrieved 2024-02-04.
- Jansana, Ramon (2022). "Algebraic Propositional Logic"↗. *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived↗ from the original on 20 December 2016. Retrieved 19 February 2024.
- Janssen, Theo M. V.; Zimmermann, Thomas Ede (2021). "Montague Semantics"↗. *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived↗ from the original on 16 April 2021. Retrieved 12 February 2024.
- Jiang, Yan (2016). "Deixis and Anaphora". *A Reference Grammar of Chinese*↗. Cambridge University Press. ISBN 978-0-521-76939-6. Retrieved 2024-02-24.

-
- Johnstone, P. T. (1987). *Notes on Logic and Set Theory*. Cambridge University Press. ISBN 978-0-521-33692-5. Retrieved 2024-02-19.
- Jun, J. S. (2009). "Lexical Conceptual Structure". In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Kay, Christian (2015). *English Historical Semantics*. Edinburgh University Press. ISBN 978-0-748-64479-7. Retrieved 2024-02-23.
- Kearns, Kate (2011). *Semantics*. Palgrave Macmillan. ISBN 978-0-333-71701-1.
- King, Jeffrey C. (2006). "Semantics". In Borchert, Donald M. (ed.). *The Encyclopedia of Philosophy*. 8: *Price - Sextus Empiricus* (2nd ed.). Thomson Gale, Macmillan Reference. ISBN 978-0-028-65788-2.
- Kortmann, Bernd (2020). *English Linguistics: Essentials*. Springer Nature. ISBN 978-3-476-05678-8. Retrieved 2024-02-25.
- Kretzmann, Norman (2006). "Semantics, History of". In Borchert, Donald M. (ed.). *The Encyclopedia of Philosophy*. 8: *Price - Sextus Empiricus* (2nd ed.). Thomson Gale, Macmillan Reference. ISBN 978-0-028-65788-2.
- Krifka, Manfred (2001). "Compositionality". In Wilson, Robert A.; Keil, Frank C. (eds.). *The MIT Encyclopedia of the Cognitive Sciences (MITECS)*. MIT Press. ISBN 978-0-262-73144-7. Retrieved 2024-02-09.
- Kuche, Louise; Rowland, Susan (2023). "Rhetoric, Influence, and Persuasion". In Rowland, Susan; Kuchel, Louise (eds.). *Teaching Science Students to Communicate: A Practical Guide*. Springer Nature. ISBN 978-3-030-91628-2. Retrieved 2024-02-24.
- L'Homme, Marie-Claude (2020). *Lexical Semantics for Terminology: An Introduction*. John Benjamins. ISBN 978-9-027-26178-6. Retrieved 2024-02-15.
- Leach, Stephen; Tartaglia, James (2018). "Postscript: The Blue Flower". *The Meaning of Life and the Great Philosophers*. Routledge. ISBN 978-1-315-38592-1.
- Leaman, Oliver (2015). *The Biographical Encyclopedia of Islamic Philosophy*. Bloomsbury. ISBN 978-1-472-56945-5. Retrieved 2024-02-23.
- Lepore, E. (2009). "Truth Conditional Semantics and Meaning". In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Lepore, Ernest; Ludwig, Kirk (2009). *Donald Davidson's Truth-Theoretic Semantics*. Clarendon. ISBN 978-0-191-53749-3. Retrieved 2024-02-23.
- Lewis, David (2012). "General Semantics". In Davidson, Donald; Harman, Gilbert (eds.). *Semantics of Natural Language*. Springer. ISBN 978-9-401-02557-7. Retrieved 2024-02-04.
- Li, Fuyin (2021). "Cognitive Semantics". *Oxford Bibliographies*. Archived from the original on 25 March 2023. Retrieved 10 February 2024.
- Löbner, Sebastian (2013). *Understanding Semantics* (2nd ed.). Routledge. ISBN 978-0-415-82673-0.
- Lyons, John (1996). *Semantics 1* (Repr. ed.). Cambridge University Press. ISBN 978-0-521-21473-5.
- Magnus, P. D.; Button, Tim; Thomas-Bolduc, Aaron; Zach, Richard; Loftis, J. Robert; Trueman, Robert (2021). *Forall X: Calgary: An Introduction to Formal Logic* (PDF). University of Calgary. ISBN 979-8-527-34950-4. Archived (PDF) from the original on 16 February 2023. Retrieved 27 March 2023.
- Malpas, Jeff (2014). "Introduction: Hermeneutics and Philosophy". In Malpas, Jeff; Gander, Hans-Helmuth (eds.). *The Routledge Companion to Hermeneutics*. Routledge. ISBN 978-1-317-67664-5. Retrieved 2024-02-23.
- Márquez, Miguel Fuster (2011). *Working with Words: An Introduction to English Linguistics*. Universitat de València. ISBN 978-8-437-08579-1. Retrieved 2024-02-15.
- Marti, Genoveva (1998). "Sense and Reference". *Routledge Encyclopedia of Philosophy*. Routledge. doi:10.4324/9780415249126-X038-1. ISBN 978-0-415-25069-6. Retrieved 2024-02-09.
- Martin, R. M. (1953). "On the Semantics of Hobbes". *Philosophy and Phenomenological Research*. **14** (2): 205–211. JSTOR 2103327.
- Meier-Oeser, Stephan (2019). "8. Meaning in Pre-19th Century Thought". *Foundations, History, and Methods*. De Gruyter Mouton. ISBN 978-3-110-37373-8.
- Meulen, Alice ter (2008). "Philosophy of Language and Linguistics". In Dascal, Marcelo; Gerhardus, Dietfried; Lorenz, Kuno; Meggle, Georg (eds.). *Sprachphilosophie [Philosophy of Language]* (in German). Walter de Gruyter. ISBN 978-3-110-20329-5. Retrieved 2024-02-24.
- Moeschler, Jacques (2007). "Introduction to Semantics". In Rajman, Martin (ed.). *Speech and Language Engineering*. EPFL Press. ISBN 978-0-824-72219-7. Retrieved 2024-02-15.
- Morris, Kevin; Preti, Consuelo (2023). *Early Analytic Philosophy: An Inclusive Reader with Commentary*. Bloomsbury. ISBN 978-1-350-32361-2. Retrieved 2024-02-18.
- Mosses, Peter D. (2003). "The Varieties of Programming Language Semantics (And Their Uses)". In Bjørner, Dines; Broy, Manfred; Zamulin, Alexandre (eds.). *Perspectives of System Informatics: 4th International Andrei Ershov Memorial Conference, PSI 2001, Akademgorodok, Novosibirsk, Russia, July 2-6, 2001, Revised Papers*. Springer. ISBN 978-3-540-45575-2. Retrieved 2024-02-19.
- Murphy, M. L.; Koskela, Anu (2010). *Key Terms in Semantics*. A&C Black. ISBN 978-1-847-06276-5. Retrieved 2024-02-09.
- Murphy, M. L. (2009). "Antonymy and Incompatibility". In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Mushayabasa, Godwin (2014). *Translation Technique in the Peshitta to Ezekiel 1-24: A Frame Semantics Approach*. Brill. ISBN 978-9-004-27443-3. Retrieved 2024-02-15.
- Nerlich, Brigitte (2019). "9. The Emergence of Linguistic Semantics in the 19th and Early 20th Century". *Foundations, History, and Methods*. De Gruyter Mouton. ISBN 978-3-110-37373-8.
- Nerlich, Brigitte (1992). *Semantic Theories in Europe, 1830–1930: From Etymology to Contextuality*. John Benjamins. ISBN 978-9-027-27726-8. Retrieved 2024-02-23.
- Noth, Winfried (1990). *Handbook of Semiotics*. Indiana University Press. ISBN 978-0-253-20959-7. Retrieved 2024-02-09.
- Nouwen, Rick; Brasoveanu, Adrian; van Eijck, Jan; Visser, Albert (2022). "Dynamic Semantics". *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived from the original on 25 February 2024. Retrieved 13 February 2024.
- Olkowski, Dorothea; Pirovolakis, Efichis (2019). *Deleuze and Guattari's Philosophy of Freedom: Freedom's Refrains*. Routledge. ISBN 978-0-429-66352-9. Retrieved 2024-02-09.
- Östman, Jan-Ola; Fried, Mirjam (2005). *Construction Grammars: Cognitive Grounding and Theoretical Extensions*. John Benjamins. ISBN 978-9-027-21823-0. Retrieved 2024-02-23.
- O'Regan, Gerard (2020). *Mathematics in Computing: An Accessible Guide to Historical, Foundational and Application Contexts*. Springer Nature. ISBN 978-3-030-34209-8. Retrieved 2024-02-19.
- Palmer, Frank Robert (1976). *Semantics: A New Outline*. Cambridge University Press. ISBN 978-0-521-20927-4.
- Park-Johnson, Sunny K.; Shin, Sarah J. (2020). *Linguistics for Language Teachers: Lessons for Classroom Practice*. Routledge, Taylor & Francis. ISBN 978-1-138-68182-8.
- Partee, Barbara (1997). "Opacity and Scope". In Ludlow, Peter (ed.). *Readings in the Philosophy of Language*. MIT Press. ISBN 978-0-262-62114-4. Retrieved 2024-02-24.
- Pavel, Thomas G. (1986). *Fictional Worlds*. Harvard University Press. ISBN 978-0-674-29966-5. Retrieved 2024-02-18.
- Pearce, Kenneth L. (2022). "Berkeley's Theory of Language". *The Oxford Handbook of Berkeley*. Oxford University Press. ISBN 978-0-190-87342-4. Retrieved 2024-02-18.
- Peeters, Bert (2006). *Semantic Primes and Universal Grammar: Empirical Evidence from the Romance Languages*. John Benjamins. ISBN 978-9-027-23091-1. Retrieved 2024-02-15.
- Pelletier, Francis Jeffry (1994). "The Principle of Semantic Compositionality". *Topoi*. **13** (1): 11–24. doi:10.1007/BF00763644.
- Pollock, John L. (2017). *The Foundations of Philosophical Semantics*. Princeton University Press. ISBN 978-1-400-88646-3. Retrieved 2024-02-15.
- Portner, Paul H.; Partee, Barbara H. (2008). *Formal Semantics: The Essential Readings*. John Wiley & Sons. ISBN 978-0-470-75818-2. Retrieved 2024-02-15.
- Provenzola, Thomas (2013). "Abelard, Peter". In Hindson, Edward E.; Mitchell, Daniel R. (eds.). *The Popular Encyclopedia of Church History: The People, Places, and Events That Shaped Christianity*. Harvest House. ISBN 978-0-736-94807-4. Retrieved 2024-02-23.
- Pustejovsky, J. (2009). "Lexical Semantics". In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Pustejovsky, J. (2006). "Lexical Semantics: Overview". *Encyclopedia of Language & Linguistics* (2nd ed.). Elsevier. ISBN 978-0-080-44854-1. Retrieved 2024-02-15.
- Reif, Monika; Polzenhagen, Frank (2023). *Cultural Linguistics and Critical Discourse Studies*. John Benjamins. ISBN 978-9-027-24952-4. Retrieved 2024-02-09.
- Riemer, Nick (2010). *Introducing Semantics*. Cambridge University Press. ISBN 978-0-521-85192-3. Retrieved 2024-02-04.
- Riemer, Nick (2016). "Internalist Semantics: Meaning, Conceptualization and Expression". In Riemer, Nick (ed.). *The Routledge Handbook of Semantics*. Routledge. ISBN 978-0-415-66173-7.
- Rooij, Robert van (2012). "Meaning and Use". In Kempson, Ruth M.; Fernando, Tim; Asher, Nicholas (eds.). *Philosophy of Linguistics*. Elsevier. ISBN 978-0-444-51747-0. Retrieved 2024-02-18.
- Rowe, Bruce M.; Levine, Diane P. (2015). *A Concise Introduction to Linguistics*. Routledge. ISBN 978-1-317-34928-0. Retrieved 2024-02-24.
- Saeed, John I. (2009). *Semantics* (3rd ed.). Wiley-Blackwell. ISBN 978-1-405-15639-4.
- Sanford, A. J. (2009). "Psychology, Semantics in". In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Seachris, Joshua. "Meaning of Life: Contemporary Analytic Perspectives". *Internet Encyclopedia of Philosophy*. Retrieved 19 September 2024.
- Shapiro, Stewart; Kouri Kissel, Teresa (2024). "Classical Logic". *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived from the original on 3 May 1998. Retrieved 19 February 2024.
- Shead, Stephen (2011). *Radical Frame Semantics and Biblical Hebrew: Exploring Lexical Semantics*. Brill. ISBN 978-9-004-22218-2. Retrieved 2024-02-15.
- Shi, Zhongzhi (2017). *Mind Computation*. World Scientific. ISBN 978-981-314-582-5. Retrieved 2024-02-19.
- Smith, Edward E.; Rips, Lance J.; Shoben, Edward J. (1975). "Semantic Memory and Psychological Semantics". In Bower, Gordon H. (ed.). *The Psychology of Learning and Motivation*. Academic Press. ISBN 978-0-080-86359-7. Retrieved 2024-02-19.
- Speaks, Jeff (2021). "Theories of Meaning". *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived from the original on 26 September 2019. Retrieved 10 February 2024.
- Staal, J. F. (1966). "Indian Semantics, I". *Journal of the American Oriental Society*. **86** (3): 304–311. ISSN 0003-0279. JSTOR 597038.
- Steinfatt, Thomas M. (2009). "General Semantics". In Littlejohn, Stephen W.; Foss, Karen A. (eds.). *Encyclopedia of Communication Theory*. Sage. ISBN 978-1-412-95937-7. Retrieved 2024-02-24.
- Strauven, Wanda (2018). "Marinetti's *Tattilismo* Revisited". In Catanese, Rossella (ed.). *Futurist Cinema: Studies on Italian Avant-garde Film*. Amsterdam University Press. ISBN 978-9-048-52523-2. Retrieved 2024-02-18.
- Sun, Zhenbin (2014). *Language, Discourse, and Praxis in Ancient China*. Springer. ISBN 978-3-642-54865-9.
- Szabó, Zoltán Gendler (2020). "Compositionality". *Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived from the original on 25 February 2024. Retrieved 7 February 2024.
- Taylor, J. R. (2009). "Cognitive Semantics". In Allan, Keith (ed.). *Concise Encyclopedia of Semantics*. Elsevier. ISBN 978-0-080-95969-6. Retrieved 2024-02-04.
- Taylor, J. R. (2013). "Cognitive Semantics and Structural Semantics". In Blank, Andreas; Koch, Peter (eds.). *Historical Semantics and Cognition*. Walter de Gruyter. ISBN 978-3-110-80419-5. Retrieved 2024-02-15.
- Taylor, John R. (2017). "Lexical Semantics". *The Cambridge Handbook of Cognitive Linguistics*. Cambridge University Press. ISBN 978-1-107-54420-8. Archived from the original on 2024-02-15. Retrieved 2024-02-15.
- Tondl, L. (2012). *Problems of Semantics: A Contribution to the Analysis of the Language Science*. Springer. ISBN 978-9-400-98364-9. Retrieved 2024-02-09.
- Trips, Carola (2009). *Lexical Semantics and Diachronic Morphology: The Development of -hood, -dom and -ship in the History of English*. Walter de Gruyter. ISBN 978-3-484-30527-4. Retrieved 2024-02-15.
- Tulving, Endel (2001). "Episodic Vs. Semantic Memory". In Wilson, Robert A.; Keil, Frank C. (eds.). *The MIT Encyclopedia of the Cognitive Sciences (MITECS)*. MIT Press. ISBN 978-0-262-73144-7. Retrieved 2024-02-19.
- Valin, Robert D. Van (2008). *Investigations of the Syntax-semantics-pragmatics Interface*. John Benjamins. ISBN 978-9-027-20572-8. Retrieved 2024-02-15.
- Vámos, Tibor (2019). "3. Knowledge Representation". In Liebowitz, Jay (ed.). *The Handbook of Applied Expert Systems*. CRC Press. ISBN 978-0-429-60697-7. Retrieved 2024-02-23.
- Whiting, Daniel. "Conceptual Role Semantics". *Internet Encyclopedia of Philosophy*. Archived from the original on 17 February 2024. Retrieved 17 February 2024.
- Wierzbicka, Anna (1988). *The Semantics of Grammar*. John Benjamins. ISBN 978-9-027-23019-5. Retrieved 2024-02-15.
- Williams, Noel (1997). "The Semantics of the Word *Fairy*: Making Meaning Out of Thin Air". In Narváez, Peter (ed.). *The Good People: New Fairylore Essays*. University Press of Kentucky. ISBN 978-0-813-10939-8. Retrieved 2024-02-10.
- Yule, George (2010). *The Study of Language* (4th ed.). Cambridge University Press. ISBN 978-0-521-76527-5.
- Zaefferer, D. (2019). "Introduction: Universals and Semantics". In Zaefferer, D. (ed.). *Semantic Universals and Universal Semantics*. Walter de Gruyter. ISBN 978-3-110-87052-7. Retrieved 2024-02-04.
- Zalta, Edward N. (2022). "Gottlob Frege". *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University. Archived from the original on 25 February 2024. Retrieved 9 February 2024.
- Zhao, Ming (2023). "Preface". *Cultural Semantics in the Lexicon of Modern Chinese*. Brill. ISBN 978-9-004-53518-3. Archived from the original on 2024-02-15. Retrieved 2024-02-15.

External links

[edit]

Wikimedia Commons has media related to **Semantics**.

Look up **semantics** in Wiktionary, the free dictionary.

Wikibooks has a book on the topic of: **Linguistics/Semantics**

 V • T • E Philosophy of language [show]
 V • T • E Formal semantics (natural language) [show]

Categories: Semantics | Concepts in logic | Grammar | Linguistics terminology | Meaning (philosophy of language)

This page was last edited on 11 July 2025, at 15:01 (UTC).

Text is available under the **Creative Commons Attribution-ShareAlike 4.0 License**; additional terms may apply. By using this site, you agree to the **Terms of Use** and **Privacy Policy**. Wikipedia® is a registered trademark of the **Wikimedia Foundation, Inc.**, a non-profit organization.

Privacy policy

About Wikipedia

Disclaimers

Contact Wikipedia

Code of Conduct

Developers

Statistics

Cookie statement

Mobile view